Report on KCC Summer Internship Programme 2026
Kalpana Chawla Centre (KCC) for Research in Space Science & Technology
Chandigarh University

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Introduction

The **Kalpana Chawla Centre (KCC) for Research in Space Science & Technology**, Chandigarh University, successfully organized the **KCC Summer Internship Programme 2026** from **15 June to 10 July 2026** under the **CU Space Technology Initiative**. The programme was conceived as a **flagship initiative** to develop a highly skilled, innovation-driven workforce capable of contributing to India's rapidly expanding space ecosystem.

With India's space sector undergoing unprecedented transformation through increased public-private participation, commercialization, and technological innovation, there is a growing need to equip students with multidisciplinary knowledge and practical skills in advanced space technologies. In response to this national need, the internship programme was designed to bridge the gap between academic learning and real-world engineering through experiential education, hands-on training, and direct interaction with experts from **ISRO, the Indian Institute of Space Science and Technology (IIST), and other organizations under the Department of Space (DoS), Government of India**.

The programme brought together **50 undergraduate and postgraduate students** from diverse engineering and science disciplines for an intensive **25-day** learning experience comprising **more than 150 hours** of expert lectures, laboratory sessions, hands-on satellite engineering activities, technical demonstrations, and multidisciplinary project-based learning. Participants received training in **Satellite Systems Engineering, Spacecraft Architecture, Embedded Systems, Artificial Intelligence for Space Applications, Remote Sensing, Ground Segment Technologies, UAV Systems, and other emerging space technologies**, thereby strengthening both their technical competence and innovation capabilities.

A defining feature of the programme was the strong collaboration established with **ISRO, IIST, IN-SPACe, URSC, VSSC, NRSC, ISTRAC, NSIL, SCL**, and other national organizations. More than **15 distinguished scientists, engineers, academicians, and industry experts** directly mentored the students, providing invaluable insights into India's space missions, satellite technologies, mission operations, and future opportunities in the space sector. In addition to mentoring students, the visiting experts also held interactions with the **University leadership and faculty members**, offering valuable guidance on strengthening Chandigarh University's capabilities in **space technology education, research, laboratory development, student satellite programmes, and strategic collaborations**.

A landmark achievement of this initiative was the **academic collaboration between the Indian Institute of Space Science and Technology (IIST) and Chandigarh University**. For the first time, **IIST partnered with the Kalpana Chawla Centre (KCC)** to conduct the **Satellite Engineering Boot Camp**, providing students with intensive hands-on training in satellite systems engineering. The successful collaboration culminated in the issuance of **joint certificates by Chandigarh University and IIST** for the Boot Camp participants, further enhancing the academic significance and national recognition of the programme.

The **KCC Summer Internship Programme 2026** marks an important milestone in Chandigarh University's vision of establishing the **Kalpana Chawla Centre** as a **Centre of Excellence in Space Science and Technology**. The programme has laid a strong foundation for future initiatives, including **CU-Sat**, **StratoSat**, advanced satellite engineering laboratories, ground station development, collaborative research, and long-term partnerships with national space organizations, thereby strengthening the University's contribution to India's growing space ecosystem.

Inaugural Ceremony

The **KCC Summer Internship Programme 2026** was formally inaugurated on **15 June 2026**, marking the launch of Chandigarh University's flagship initiative under the **CU Space Technology Initiative**. The programme was inaugurated by **Shri Sudhakar, Distinguished Scientist and Deputy Director, U R Rao Satellite Centre (URSC), ISRO**, who graced the occasion as the **Chief Guest**, in the esteemed presence of the **Hon'ble Vice Chancellor, Prof. (Dr.) Raviraja N. Seetharam**. The inauguration reflected the University's growing commitment to advancing education, research, and innovation in space science and technology through strategic collaborations with national space organizations.

The inaugural ceremony was attended by eminent scientists and experts from **ISRO**, the **Indian Institute of Space Science and Technology (IIST)**, and the **Semiconductor Laboratory (SCL), Government of India**, demonstrating the strong academic and technical partnerships established between Chandigarh University and India's premier space and strategic technology institutions.

The distinguished guests included:

- **Shri Sudhakar**, Distinguished Scientist & Deputy Director, **U R Rao Satellite Centre (URSC), ISRO** *(Chief Guest)*
- **Shri Jyothi Soman**, Group Director, **U R Rao Satellite Centre (URSC), ISRO**
- **Dr. Kamaljeet Singh**, Director General, **Semiconductor Laboratory (SCL), Government of India**
- **Prof. Priyadarshnam**, Professor of Avionics & Head, **Small Satellite Payload Space Research Centre (SSPACE), Indian Institute of Space Science and Technology (IIST)**

- **Er. Abhishek Verma**, **Indian Institute of Space Science and Technology (IIST)**

During the inaugural session, the distinguished speakers highlighted the rapid growth of India's space sector and emphasized the importance of nurturing a highly skilled workforce through **experiential learning, multidisciplinary education, innovation, and strong academia–industry collaboration**. They encouraged students to develop technical excellence, embrace emerging technologies, and actively contribute to India's future space missions and the expanding national space ecosystem.

The inauguration also marked the commencement of the **Satellite Engineering Boot Camp**, conducted with the academic support of the **Indian Institute of Space Science and Technology (IIST)**. The Boot Camp provided students with intensive hands-on training in satellite systems engineering, subsystem design, integration, and mission-oriented engineering practices. This collaborative initiative laid the foundation for the intensive **25-day internship programme**, setting the stage for an enriching learning experience that combined expert mentorship, laboratory training, and multidisciplinary project-based learning.

The inaugural ceremony reaffirmed Chandigarh University's commitment to establishing the **Kalpana Chawla Centre (KCC)** as a leading **Centre of Excellence in Space Science, Satellite Engineering, Research, Innovation, and Capacity Building**, dedicated to developing the next generation of scientists, engineers, and technology leaders for India's growing space sector.

Programme Highlights

The **KCC Summer Internship Programme 2026** was designed as an intensive, multidisciplinary training programme that combined academic learning with practical engineering experience. Conducted over **25 days**, the programme provided students with direct exposure to satellite engineering, space technologies, emerging research areas, and interactions with leading scientists from India's premier space organizations. Through expert lectures, laboratory sessions, hands-on training, and project-based learning, the internship successfully created a unique platform for developing future-ready professionals for the Indian space sector.

Programme Component	Details
Programme Duration	**25 Days**
Programme Period	**15 June – 10 July 2026**
Students Trained	**50 Undergraduate & Postgraduate Students**
Training Delivered	**More than 150 Hours**
Expert Sessions	**15+ Distinguished Scientists, Engineers & Industry Experts**
Flagship Activity	**Satellite Engineering Boot Camp** in collaboration with **IIST**

Programme Component	Details
Training Methodology	Expert Lectures, Laboratory Sessions, Hands-on Experiments, Technical Demonstrations, Project-Based Learning, Team Activities
Technology Domains Covered	Satellite Systems Engineering, Spacecraft Architecture, Embedded Systems, Artificial Intelligence, Remote Sensing & GIS, UAV Technologies, Ground Segment & TT&C, Mission Planning, Space Applications
Major Collaborating Organizations	**ISRO, IIST, URSC, VSSC, NRSC, ISTRAC, IN-SPACe, NSIL, SCL**
Certification	**Certificate of Participation by Chandigarh University** and **Joint Certificate with IIST** for the Satellite Engineering Boot Camp participants

Key Highlights

- Successfully organized the **first KCC Summer Internship Programme** under the **CU Space Technology Initiative**.
- Conducted the **first Satellite Engineering Boot Camp** at Chandigarh University with academic support from **IIST**.
- Facilitated direct interaction between students and **15+ eminent scientists and experts** from India's leading space organizations.
- Delivered intensive **hands-on training** in satellite systems engineering and emerging space technologies.
- Strengthened strategic collaborations between **Chandigarh University**, **ISRO**, **IIST**, and other organizations under the **Department of Space (DoS)**.
- Established the foundation for future initiatives, including **CU-Sat**, **StratoSat**, advanced satellite engineering laboratories, ground station development, and collaborative research programmes.

Academic and Technical Collaboration

One of the most significant achievements of the **KCC Summer Internship Programme 2026** was the establishment of strong academic and technical collaborations with India's premier space organizations, research institutions, and strategic technology agencies. These collaborations enabled Chandigarh University to provide students with direct exposure to national-level expertise, emerging space technologies, and mission-oriented engineering practices.

The internship programme was conducted with the active participation and academic support of distinguished scientists, engineers, academicians, and experts from organizations under the **Department of Space (DoS), Government of India**, including:

- Indian Space Research Organisation (ISRO)
- Indian Institute of Space Science and Technology (IIST)
- U R Rao Satellite Centre (URSC)
- Vikram Sarabhai Space Centre (VSSC)
- National Remote Sensing Centre (NRSC)
- ISRO Telemetry, Tracking and Command Network (ISTRAC)
- Indian National Space Promotion and Authorization Centre (IN-SPACe)
- NewSpace India Limited (NSIL)
- Semiconductor Laboratory (SCL), Government of India
- Leading Space Industry Experts

Throughout the programme, **more than 15 distinguished scientists, engineers, and academicians** delivered expert lectures, conducted technical demonstrations, mentored student projects, and interacted closely with students, faculty members, and the University leadership. These interactions provided valuable insights into India's space missions, satellite technologies, launch vehicles, mission operations, remote sensing, satellite communications, quality assurance, semiconductor technologies, and the future of the Indian space sector.

A landmark achievement of the programme was the **academic partnership established with the Indian Institute of Space Science and Technology (IIST)**. For the first time, **IIST collaborated with Chandigarh University through the Kalpana Chawla Centre (KCC)** to organize the **Satellite Engineering Boot Camp**, enabling students to receive intensive hands-on training in satellite systems engineering under the guidance of **Prof. Priyadarshnam** and the **SSPACE, IIST** team. This collaboration was further strengthened through the issuance of **joint certificates** for the Boot Camp participants, recognizing the shared academic efforts of **Chandigarh University** and **IIST**.

Beyond student training, the visiting experts also held productive interactions with the **University leadership and faculty members**, sharing valuable recommendations on strengthening Chandigarh University's capabilities in **space technology education, advanced laboratory development, student satellite programmes, collaborative research, remote sensing applications, artificial intelligence, and long-term institutional partnerships**.

These collaborations have significantly enhanced the academic ecosystem of the **Kalpana Chawla Centre** and established a strong foundation for future initiatives, including **CU-Sat**, **StratoSat**, advanced satellite engineering laboratories, ground station development, faculty development programmes, joint research projects, and strategic partnerships with national and international space organizations.

Satellite Engineering Boot Camp

The **Satellite Engineering Boot Camp** was the flagship component of the **KCC Summer Internship Programme 2026**, providing students with an immersive, hands-on learning experience in the design, development, and integration of satellite systems.

The Boot Camp was conducted in academic collaboration with the **Indian Institute of Space Science and Technology (IIST), Thiruvananthapuram**, under the leadership of **Prof. Priyadarshnam**, Professor of Avionics & Head, **Small Satellite Payload Space Research Centre (SSPACE)**, with the active support of **Er. Abhishek Verma** and the IIST team. This collaboration marked the **first joint academic initiative between Chandigarh University and IIST** in the field of satellite engineering and space systems education.

Unlike conventional classroom-based training, the Boot Camp adopted a **systems engineering approach**, enabling students to understand the complete lifecycle of satellite development—from mission concept and system requirements to subsystem design, integration, testing, and mission operations. Through expert guidance and practical demonstrations, participants gained first-hand exposure to the engineering principles employed in the development of modern small satellites.

The Boot Camp covered a wide range of topics, including:

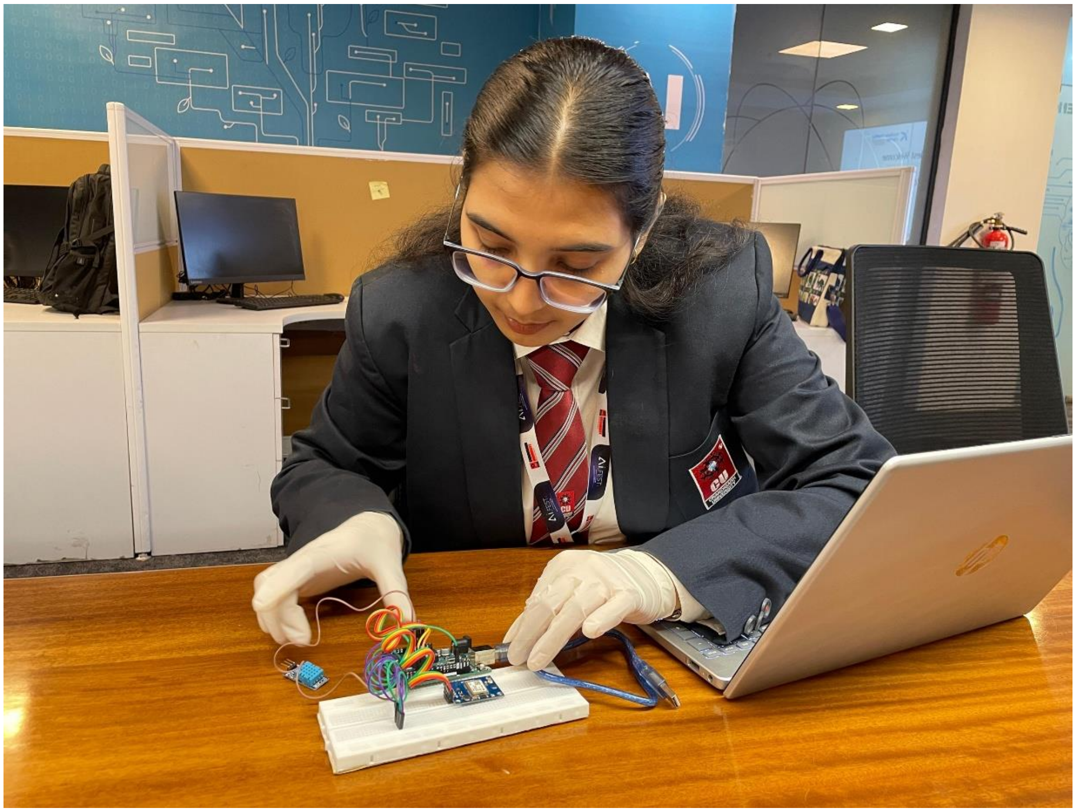
- Mission Analysis and Systems Engineering
- Satellite Bus Architecture
- Payload Selection and Integration
- Structural and Thermal Design
- Electrical Power Systems (EPS)
- On-board Computer (OBC) and Flight Software
- Communication Systems and TT&C
- Attitude Determination and Control Systems (ADCS)
- Sensors, GPS, and Embedded Systems
- Antenna Design and RF Communication
- Ground Station Development
- Software Defined Radio (SDR) and GNU Radio
- Mission Planning, Integration, Testing, and Mission Operations

Students actively participated in hands-on laboratory sessions, subsystem-level demonstrations, and multidisciplinary team activities, enabling them to translate theoretical concepts into practical engineering solutions. The Boot Camp also fostered collaborative problem-solving, systems thinking, and engineering design skills essential for future careers in the space sector.

A significant outcome of this initiative was the issuance of **joint certificates by Chandigarh University and the Indian Institute of Space Science and Technology (IIST)** for students who successfully completed the Satellite Engineering Boot Camp. The approval granted by the **Hon'ble Vice Chancellor, IIST**, to include the **IIST logo** on the certificates further strengthened the academic partnership between the two institutions and added significant value to the students' professional credentials.

The successful execution of the **Satellite Engineering Boot Camp** established a strong foundation for future collaborative activities between **Chandigarh University** and **IIST**, including student satellite development, faculty development programmes, joint research initiatives, advanced laboratory development, and future capacity-building programmes in space science and satellite engineering.

(Add some photos of Boot Camp)



Programme Outcomes

The KCC Summer Internship Programme successfully achieved its objectives by:

- Providing practical exposure to **Satellite Systems Engineering** and spacecraft technologies.
- Developing competencies in systems engineering, payload integration, embedded systems, mission planning, remote sensing, AI, and ground systems.
- Enhancing technical capabilities through laboratory-based experiential learning.
- Promoting multidisciplinary teamwork and innovation.
- Strengthening research aptitude among undergraduate and postgraduate students.
- Establishing strong collaborations between Chandigarh University and India's leading space organizations.
- Creating a strong foundation for future initiatives such as **CU-Sat**, **StratoSat**, advanced Ground Station development, and student-led satellite missions.

Joint Certification with IIST

One of the significant achievements of the programme was the issuance of **Joint Certificates** for the participants of the **Satellite Engineering Boot Camp**.

With the approval of the **Hon'ble Vice Chancellor, Indian Institute of Space Science and Technology (IIST)**, the certificates carry the **IIST logo** in recognition of the academic collaboration and the valuable contribution of **Prof. Priyadarshnam** and the **SSPACE, IIST** team.

This joint certification significantly enhances the academic value of the programme and reflects the growing partnership between **Chandigarh University** and **IIST** in promoting excellence in space science and satellite engineering education.

Impact of the Programme

The KCC Summer Internship Programme has successfully established a vibrant ecosystem for space technology education at Chandigarh University by:

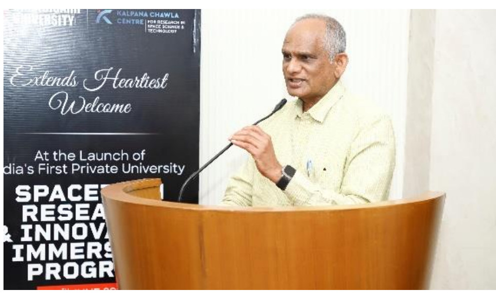
- Creating opportunities for direct interaction with eminent ISRO scientists.
- Promoting experiential and project-based learning.
- Strengthening industry-academia-government collaboration.
- Inspiring students to pursue careers in India's expanding space sector.
- Positioning the Kalpana Chawla Centre as an emerging national hub for Space Science and Technology education.

Valedictory Function

The **Valedictory Function** of the **KCC Summer Internship Programme 2026** was held on **10 July 2026** in the distinguished presence of **Dr. M. Nageswara Rao**, Distinguished Scientist and former Satish Dhawan Professor Scientist, ISRO, who graced the occasion as the **Chief Guest**. The function was attended by the Hon'ble Vice Chancellor, senior university leadership, faculty members, distinguished scientists, industry experts, and participating students.

During the programme, the achievements of the internship were presented, student projects and learning outcomes were showcased, and certificates were distributed to all participants. In his address, Dr. M. Nageswara Rao appreciated Chandigarh University's visionary initiative in establishing the Kalpana Chawla Centre as an emerging hub for space science and technology education and encouraged students to pursue careers in satellite engineering, research, innovation, and India's growing space sector.

The valedictory ceremony marked the successful completion of the University's inaugural flagship Space Technology Internship Programme and reaffirmed Chandigarh University's commitment to developing the next generation of scientists, engineers, innovators, and technology leaders.



Conclusion

The successful completion of the **KCC Summer Internship Programme 2026** marks a significant milestone in Chandigarh University's vision of establishing the **Kalpana Chawla Centre (KCC)** as a **Centre of Excellence in Space Science, Satellite Engineering, Research, Innovation, and Capacity Building**.

Beyond providing high-quality technical training, the programme established meaningful collaborations with **ISRO, IIST, IN-SPACe, URSC, VSSC, NRSC, ISTRAC, NSIL, SCL**, and other leading organizations, creating a unique platform where students learned directly from India's foremost space scientists and engineers.

The internship has laid a strong foundation for future initiatives, including **CU-Sat**, **StratoSat**, advanced satellite engineering laboratories, Ground Station development, collaborative research, and innovation-driven projects that will further strengthen Chandigarh University's contribution to India's growing space ecosystem.

The **Kalpana Chawla Centre** remains committed to nurturing future scientists, engineers, innovators, and entrepreneurs who will contribute to India's vision of becoming a global leader in **Space Science and Technology**.

Submitted by

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